

Is Time-Zone Proximity an Advantage for Software Development? The Case of the Brazilian IT Industry

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Abstract—Brazil has been emerging as a destination for IT software and services. The country already had a strong domestic base of IT clients to global companies. One of the competitive factors is time zone location. Brazil has positioned itself as easy for collaboration because of time zone overlap with its primary partners in North America and Europe. In this paper we examine whether time zone proximity is an advantage for software development by conducting a country-level field study of the Brazilian IT industry using a cross section of firms. The results provide some support for the claims of proximity benefits. The Brazil-North dyads use moderate timeshifting that is perceived as comfortable for both sides. The voice coordination that the time overlap permits helps address coordination challenges and foster relationships. One company, in particular, practiced such intense time zone aligned collaboration using agile methods that we labeled this Real-time Simulated Co-location.

Index Terms—Time-zone proximity, Brazilian IT industry, Globalization of IT

I. INTRODUCTION

With its political stability, talent pool and reliable telecommunications infrastructure, Brazil has been emerging as a key destination for IT software and services [1]. Brazil's IT industry is large, and the sector employs 1.7 million people, including programmers, systems analysts and managers. Brazil's IT-BPO export market turned over US\$ 2.4 billion in 2010 (including IT-BPO services captive centers and software).

Yet Brazil is still a small global player. India is the largest competitor in the new "offshoring industry," exporting at 20 times Brazil. Brazil has a strong domestic base of IT clients and for this reason the vast majority of the Brazilian companies are focused on domestic clients and do not export. But this is changing. Brazil is becoming a world player in the IT-BPO area. As stated in an At Kearney report [2], "Brazil-based IT companies such as Politec, Stefanini, and Ci&T are equal in size to many key international competitors and are successfully expanding their capabilities beyond Brazil". The formation of Brasscom, the Brazilian Association of Information Technology and Communication Companies confirms the strategic intent to play the offshoring game. The Brazilian IT industry has grown 43% since 2008 and Brasscom estimates that in 2020 Brazil will have a USD 200 billion IT industry and will export USD 20 billion [3].

Brazil is just beginning to emerge as a global IT competitor many years after India's emergence. National competitiveness

has long been of interest. Heeks et al [4] presented the success factors for nations to successfully export IT services. Factors include the familiar, such as low wages; and the less intuitive, such as intellectual property regimes. Absent in this and early lists was location. Countries have historically competed or failed because of their location [5]. But it is only recently that economic geographers have come to realize that there is a dimension of location that was previously overlooked: time zone [6].

The importance of time zones is also present in the IT industry. For example, according to Brasscom, Brazil's value proposition is based on seven pillars [3]. The first six are predictable: industry and business knowledge, qualified human resources, infrastructure, governmental support, favorable economic, political, and legal environment, cultural compatibility; but the seventh is time zone proximity.

Interestingly, nearly 100% of Brazil's IT/software export clients have time zone overlap with Brazil [3]. US companies are the main clients, accounting for over 80% of demand, followed by Latin America (especially Argentina, Chile, Colombia and Mexico), and Europe (especially Germany, Spain, France, England and Portugal).

For this reason, in this paper we investigate if time zone proximity may be advantageous. Instead of examining the time separated dyad of US-India, as Carmel did [7], we examine the North-South configuration of Brazilian software engineers with their primary partners in North America and in Europe. We intend to answer the following research question: "Are their advantages of Brazil in terms of time zone proximity and how are they manifested?" Our qualitative exploratory study shows that Brazil benefits from time zone overlap in this North-South configuration. It is an overlap of convenience for both sides.

The remainder of this paper is structured as follows. Section II presents the background on time-zone coordination. Section III presents information about the Brazilian IT industry. Section IV presents the research methodology and the study. In Section V we present the results from the study, and Section VI concludes the paper.

II. TIME-ZONE COORDINATION

Until the 1980s coordinating knowledge work was affected little by time zone differences. The diffusion of new technologies has permanently changed that. Until the fax machine most work products could not be transferred instantly

across time zones. Instead, computer tapes, or printed pages, were sent by air delivery. Coordination between distant sites was, at best, loosely coupled. Today, the picture has flipped entirely: shipping digital goods is very inexpensive and essentially instantaneous. And early in the era of globalized knowledge work, in the 1990s, we all bought into the notion, captured by Cairncross, that distance is dead [8].

As a result, since distance and time zone separation are tightly entwined in our minds, collectively we have also come to believe that temporality is dead. Indian software developers work while their American counterparts sleep and vice versa. But, as most software engineers can attest, synchronized coordination is often a necessity, for a myriad of reasons: for resolving miscommunications, for converging on difficult problems, for visualizing complex domains, for social team building and trust-building, and for creative dialogue. Therefore time zone proximity is useful [9].

Time zone differences arise when members of a dispersed team are located in multiple time zones. It has been referred to as temporal dispersion [10], temporal distance [11], and time separation [12]. As task times become more compressed and time pressures increase to meet market demands and competitive needs, coordination across time zones becomes a greater problem [12]. Coordination across time zones is inherently not solvable because we human beings tend to work during the daylight hours. The use of technology such as video-conferencing cannot overcome this fundamental problem. The time zone problem can be mitigated, but it cannot be eliminated. Time differences have become the principal obstacle to efficient coordination [8]. Today, differences in time zones are more difficult to overcome than distance separation. As task times become more compressed and time pressures increase to meet market demands and competitive needs, coordination across time zones becomes a greater problem.

The tactics that have been proposed over the years to overcome the (distance and) time-zone challenges can vary from structured to organic, and implicit. Structured means the codification of routines and tasks, so that the dispersed individuals can coordinate without having to communicate with one another (also called Mechanistic). Organic is when individuals communicate on their own, unscheduled and unstructured, in order to problem solve (ex: walking over to a colleague to speak to him/her). Implicit is related to shared mental model; shared task knowledge; knowledge of who knows what (transactive memory); knowledge of who is around and where; knowledge of who has done what and when (task awareness).

In software development, there is a tension between the organic and the structured. Structured processes, often embedded into the software development suites, are a perennial goal. But structured coordination is not enough. While software processes have become better structured, the need for organic coordination remains -- this is better facilitated by time zone proximity. Organic communication requires timeshifting when software units are very far apart but can be done easily when time zones overlap.

III. THE BRAZILIAN IT INDUSTRY

Brazil's IT industry is large: A.T. Kearney, a consultancy, estimates that the sector employs 1.7 million people, including programmers, systems analysts and managers. Brazil's IT-BPO export market turned over US\$ 2.2 billion in 2008 (including IT-BPO services captive centers and software). The Brazilian industry is growing by 6.5% a year on average since 2005.

India is the largest competitor in the new "offshoring industry," exporting at 20 times Brazil. In 2008, there were approximately 7,800 software and IT firms in Brazil, but the vast majority of these firms are focused on domestic clients and do not export.

Interestingly, nearly 100% of Brazil's IT/software export clients have time zone overlap with Brazil. Only 1.5% of the Brazil IT's clients reside outside of Brazil's overlapping time zones. US companies are the main clients, accounting for over 80% of demand, followed by Latin America (especially Argentina, Chile, Colombia and Mexico), and Europe (especially Germany, Spain, France, England and Portugal).

When India began to emerge as the IT giant, one of its competitive appeals was time zone. The authors met many global users that tell a story that goes roughly like this: "I sent them instructions in the afternoon and when I showed up the next morning it was all done." On the other hand, some tasks, especially high-end tasks, are unstructured tasks that call for planned and unplanned coordination activities. In such cases, those coordination activities that are asynchronous, like e-mail, are less effective. Therefore, software professionals have to resort to synchronous communication, which is usually a voice conversation or instant message dialogue.

Since the Brazilian IT industry is a follower in global competition, Brazilian firms and Brazilian locations often find themselves competing against Indian locations. In order to create differentiation, many Brazilian firms tout Brazil's time zone position as an advantage. For example, Ci&T¹ one of Brazil's largest IT firms, defines itself as a nearshore technology company. E-core², another Brazilian IT company, explicitly says that time zone proximity is important for more complex outsourcing engagements such as software development, and that Brazil's location benefits international projects.

IV. RESEARCH METHODOLOGY AND STUDY SETTINGS

The focus of our study was on Brazilian firms because of the time zone location. Brazilian firms tend to work with American and European companies. Brazil has separation of 1 to 6 hours with these locations depending on location and season (the seasonal/equatorial effects on daylight savings time (DST) is a two hour change). The overlap with the United States ranges from proximity with New York of one hour during Brazil's winter (Table I), and stretching all the way to a six-hour difference with San Francisco during Brazil's summer (Table II). The seasonal time zone chart does not reflect short transition periods in spring and fall seasons.

¹ <http://www.ciandt.com/us-en/value-proposition..>

² <http://www.ecore.com.br/portal/en/nearshore-outsourcing/why-brazil>

TABLE I. DURING BRAZIL'S WINTER

	4	3	2	1	Brazil	1	2	3	4	5
	US Pacific	US Mountain	US Central	US East coast					London, Lisbon	Western Europe
	GMT -7		GMT-5	GMT-4	GMT -3				GMT +1	GMT +2

TABLE II. DURING BRAZIL'S SUMMER

6 US Pacific	5 US Mountain	4 US Central	3 US East coast	2	1	Brazil	1	2	3	4	5
								London, Lisbon	Western Europe		
GMT -8	GMT -7	GMT -6	GMT-5			GMT -2		GMT 0	GMT +1		

The study team conducted 46 interviews at 15 Brazilian sites in 2010 in Rio de Janeiro, São Paulo, and Porto Alegre, including Brazilian companies, Brazilian companies acquired by foreign multinationals (MNCs), and captive centers of MNCs (see Table III). The companies/firms were selected looking for a balanced set of data, which included 9 Brazilian companies and 6 foreign multinational companies. In all cases the companies were predominantly IT/software although some did BPO. The study team also conducted interviews with 3 clients or foreign partners from the US, Portugal, and India.

All interviewees were asked questions from a semi-structured research instrument. The research team consisted of the authors, an American and a Brazilian researcher. All interviews were conducted in English, for a total of 33 hours of conversation. The two researchers took individual notes. After each day of interviews, the individual notes were grouped and then codified, generating a summary for each company.

TABLE III. COMPANIES IN THE STUDY

Firm	# People interviewed / # Hours	Type	Focus of Brazilian operations	Size of global firm (employees)
A	11 / 5	MNC	R&D	Very large
B	2 / 2	Brazilian	IT/Software	Medium
C	2 / 2	Brazilian	IT/Software	Medium
D	1 / 1	MNC	R&D	Large
E	4 / 3	MNC	IT/Software	Large
F	2 / 1	Brazilian	IT/Software	Large
G	1 / 1	Brazilian	R&D	Small
H	5 / 4	MNC	R&D	Very large
I	10 / 7	MNC	IT/Software	Very large
J	2 / 2	MNC	IT/Software	Medium
K	1 / 1	Brazilian	IT/Software	Small
L	1 / 1	Brazilian	R&D	Small
M	1 / 1	Brazilian	IT/Software	Small
N	2 / 1	Brazilian	IT/Software	Large
O	1 / 1	Brazilian	IT/Software	Small

The study covered a sample of 22 projects (see Table IV), which together illustrate the story of Brazilian collaboration.

The US (and occasionally Canada) is the main partner. MNCs have more complex project configurations, also interacting with Asia, increasing the challenges and coordination costs.

TABLE IV. CONFIGURATION OF PROJECTS

Type of company	Place of interaction				
	US / Canada only	Europe only	US / Canada, Europe	US / Canada, Europe, Asia	Europe, Asia
MNC	5		2	6	
Brazilian	4	2	2		1
Total	9	2	4	6	1
%	41%	9%	18%	27%	5%

Brazilian companies are less likely to be in complex multi-continent interactions. The figures show the interaction within projects: two multinational companies (Fig. 1 and Fig. 2) and two Brazilian companies (Fig. 3 and Fig. 4).



Fig. 1. Project configuration at a Multinational company (the Brazilian site is the hub, with the main collaboration between Brazil and Eastern Europe).

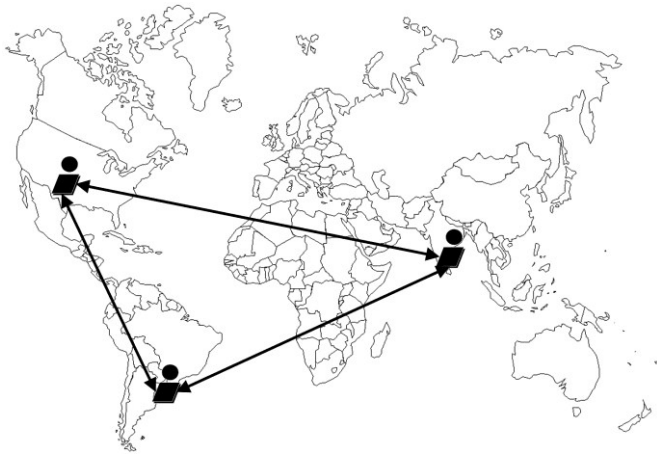


Fig. 2. Project configuration at a Multinational company.



Fig. 3. Project configuration at a Brazilian company.



Fig. 4. Project configuration at a Brazilian company.

In figure 1 the Brazilian site is the hub, with the main collaboration between Brazil and eastern Europe. Multinationals Corporations (MNCs) have more complex projects, also interacting with Asia, increasing the challenges and coordination costs. Brazilian companies are less likely to be in complex multi-continent interactions.

V. FINDINGS

Several practices were found and we begin with the most fundamental of them: timeshifting.

A. Timeshifting

Timeshifting means moving one's work hours to accommodate someone else's work hours. For example, in India, many Indian software professionals regularly stay late at the office to overlap with the US workday. In any global collaboration timeshifting is prevalent though its extent varies. Brazilian software developers need to timeshift somewhat to extend the overlap time or even create complete overlap. Table V presents the Brazilian work elements that enable timeshifting.

In Brazil, most of organizations do only modest timeshifting. 72% in the study did little, occasional or, ad-hoc timeshifting. 9% did no timeshifting. Many respondents said, in effect, that the natural overlap was sufficient. However, 19% of the firms did use systematic timeshifting, meaning that the entire Brazilian project team timeshifted daily.

For Brazilian sites there is always some time zone overlap with their partners, but this varies by location and season. The smallest gap is during the Brazilian winter when US east coast is -1h due to Daylight Savings Time. The largest gap is during the Brazilian summer when the US west coast is 6 hours away. Such a range represents calls for different timeshifting tactics.

On one hand, one manager said, "Time zone differences with the US [east and center] are small so the inconvenience and changes are not dramatic." But another said that the US West coast requires "mind-set changing when it moves back 2 hours [during the Brazilian summer]."

Generally, the Brazilian sites and the Brazilian individuals are the ones that timeshift. The Brazilians need to adjust in order to align to the headquarters/center, or to the client's hours. Time slots are dictated by the headquarters or the client; this is the norm in all global collaborations. On occasion, Brazilians wake up early to have meetings or supporting activities with Europe (03:00 or 04:00 Brazilian time), or stay until late at night to talk to the western USA.

Yet, there were a number of configurations in which individuals in the US or Europe would adjust and timeshift to Brazil's time, sometimes via teleworking. Here are two cases from the study. At one American-Brazilian collaboration (US Mountain time; 3 hours difference during Brazil's winter), the American engineers, who liked to start their day early (at 07:00 local time) expanded the overlap window with Brazil. The second case was a Portugal-Brazilian collaboration. The Portuguese didn't mind staying late at the office to work with Brazil. They thought it is easier to extend their day instead of making the Brazilians wake up earlier.

Brazilians seem to be less tolerant of timeshifting than other tech countries. They like traditional work hours and once they're done, they go home. The authors heard this sentiment at a number of Brazilian sites. In fact there was concern that frequent timeshifting, would cause employees to quit. Separately, we heard from both the Brazilians and their North American partners that the time overlap with Brazil was more

pleasant than the odd-hours needed for collaboration with Asia. From the North American partners, we also heard that the overlap when working with Brazil better their work-life balance relative to Asian collaboration.

TABLE V. WORK ELEMENTS THAT ENABLE TIMESHIFTING

Characteristic which leads to greater time zone overlap	Definition / interpretation	The Brazilian IT sector
Flextime	Employees work on flexible schedules.	Brazilian firms give employees some flexibility. MNCs give more.
Overtime work	Captures time zone overlap with extra hours.	As in the tech sector in much of the world, Brazilian tech workers work overtime, but this does not appear to be as much as in Asia.
Weekend work	Captures time zone overlap with flexible hours	Weekend work is rare
Telework	Captures time zone overlap with flexible hours from home.	Telework is still at low levels and seems to encounter resistance from both managers and employees. Foreign clients may restrict, by contract, offsite work for security reasons.
Unrestricted work regulations	Employers have no external restrictions on setting work hours or overtime expectations.	Labor laws place some restrictions on overtime.
Long workday / workweek	Capture time zone overlap at start or end point.	Some, but not all, Brazilian companies work a standard 44h workweek.
Cultural flexibility to work in nontraditional time	Employees are willing -- or even prefer -- to scatter their work hours or to timeshift.	Brazilians like to work during traditional hours. They are flexible, but they don't like dramatic timeshifting or teleworking.

We found that timeshifting varies on many factors:

- Variations within a firm. Some teams timeshift more than others.
- Variations within a team. Some individuals timeshift more than others.
- Timeshifting varies by season because of the two-hour adjustments of North-South DST.
- Operational timeshifting versus development. In some companies various support and infrastructure units regularly assign one or two employees to work in nontraditional shifts.

We also note three specific timeshifting tactics: Liaisons, telework, and overtime. A liaison is the person who bridges locations, time zones, and culture, typically a mid-level manager, such as a project manager. All the Brazilian units in this study have liaisons in spite of the time zone overlap. Telework (working from home) gives the firm greater time zone reach. However, there is little Brazilian telework and there some management resistance. One obstacle may be moonlighting: local managers want to keep an eye on their employees since some IT engineers have side projects. On the other hand, Brazilians are increasingly dealing with teleworkers abroad. This seems to benefit the clustered Brazilian sites because the isolated foreign teleworker is more likely to timeshift. Overtime also gives the firm greater time zone reach. But since Brazilian labor laws are somewhat restrictive this affects the culture of overtime relative to the more overtime-prone tech communities of India, China, and the USA.

B. Software Development Methodologies

Since the challenge of time zone and distance is work coordination, the choice of work methodology is important. As is the case globally, the choice often boils down to waterfall versus agile along with all the hybrids and derivatives of these two. Interestingly, at all the Brazilian units in this study, developers are either using or "moving to" agile methodologies as the software development methodology. That is, in some companies, only some agile practices are used, while in others agile is used intensely. We heard a number of times in our interviews that the use of agile methodologies contributes to solving the coordination and communication difficulties with global projects, including the improvement of team communication. Time zone overlap permits some key agile practices: daily stand-up meetings and paired programming. From a technical point of view, some tactics that companies used for distributed agile include: continuous integration and continuous delivery, regular (sometimes daily) builds to increase the feedback loop, test automation / test-driven development, and teams separated by functionality.

Especially noteworthy is one MNC that is an agile leader -- using all the agile practices in its collaboration between its Brazilians sites and North American sites. The project team was implementing daily continuous integration to minimize coordination problems. This approach enables frequent automated integration of code across various locations and every distributed team could work on a testable and integrated piece of code. Additional practices experienced by the project

team were: pair programming over Skype and shared desktop; daily meetings using video conference; and remote planning sections using online management tools.

C. Task Allocation

Task allocation is one of the most interesting challenges in Global Software Engineering [13], and it is a difficult trade-off for all global collaboration. When working across sites, two or more collaborating sites can choose along a dependency spectrum where they either eliminate all dependencies as much as possible or work very closely with high dependencies. We found a range of approaches in our sample. Here are some illustrations in our study, all about MNC with Brazilian sites:

- Firm 1. The number of sites working on the same project at the same time was intentionally reduced.
- Firm 2. The individuals involved in global collaboration were far-flung, highly dispersed. There was little deliberate task allocation by location. Rather, it was based on individuals and their availability of expertise.
- Firm 3. There was a clear philosophy that “We don’t break software components [across sites and time zones].”
- Firm 4. We heard about the learning progression. The executive interviewed explained that in the beginning projects were allocated to teams/sites based on where team members were located even across several time zones. Because of the high coordination, the company created centers of excellence and determined not to have more than 2 sites working on the same project.

Time-zone overlap allows work on relatively more complex tasks and projects. Some of the interviewees felt that this was an advantage for them – that their responsibility for complex projects and tasks was increased due to the time zone proximity.

D. Technological Solutions

There are no miracle technologies to overcome time zone differences. However, as indicated in the previous section, there are impressive technologies to reduce distance.

- Technological solutions for communication. Brazilian and foreign sites make heavy use of always-on IM for the day-by-day communication. Other synchronous communication tools were used frequently such as VoIP (e.g., Skype) and even plain old telephone/ mobile. Video is used in some companies for remote meetings. Telepresence (high end HD video-conferencing) is used in one company. We did not hear of Social Media (SM) tools being used as communication. Some companies do have SM tools (internal Twitter-like, internal Facebook-like), but they are not used.

- Technological solutions for collaboration. Companies use screen sharing, project management and Configuration Management tools (open source in some cases), global repositories (e.g. SharePoint), continuous integration and bug reporting tools, and wikis. Technologies to support meetings are now commonplace and global collaborators are increasingly using tools that facilitate scheduling meetings over different time zones.

- Technological solutions for awareness. Companies use always-on video, IM online status, and awareness apps integrated into their email tool.

The Brazilian companies we studied used intensive technological solutions for communication, but generally appeared less advanced in solutions for collaboration and awareness.

E. Intense Collaboration and Real-time Simulated Co-location

What is the difference between the international collaboration at different Brazilian units? After all, due to the time zone overlap, all projects have at least some real-time interaction. So, what is the difference between characteristics of communication from one Brazilian unit to another? They do not collaborate with their distant partners equally. The gradations of collaboration that we found in Brazilian units led us to discern four levels of collaboration intensity. Therefore, we begin by introducing the terminology and the four-level collaboration framework.

The study data show that Brazilian units use a range of interactive collaboration. This continuum can be categorized into 4 levels organized as the four-level framework:

Level 1

Sites rely mostly on asynchronous tools, not for cost reasons only, but due to traditional organizational mindset. Sporadic use of common synchronous tools such as IM (instant messaging), Skype (voice over IP), remote desktop, and video.

Level 2

Sites commonly use IM, status indicators, and screen sharing. Tool portfolio is still reliant more on standard technologies. Video is used for formal meetings, but is not a de facto technology.

Level 3

Sites have state-of-the-art collaborative technology to support distributed teams and far-flung individuals. Awareness tools and status indicators are heavily used, especially IM/chat.

Level 4

Sites simulate co-located environments to maximum possible. We label this as RTSC (Real-Time Simulated Co-location). This is a same-time environment with rich communication technologies. Awareness technologies usage is very high.

Most of the Brazilian companies are at Level 1, partially due to a traditional mindset. The foreign MNCs tend to be at higher levels. We visited one company that maximized its use of time zone proximity using RTSC (Real-Time Simulated Co-location). There was near complete time zone overlap, hence the “real-time.” And, via technologies, methodologies and work culture, there was a simulated feel of co-location. The collaborating sites were connected via always-on high-resolution video displays (which is most effective for a team room work environment of agile software development). The

video amplifies the awareness as if there is physical co-location. For example, as described to us, at the Brazilian site they can see when someone comes to work at the US site and the American person can notice when someone at the Brazilian site is about to pick up the phone. Awareness technologies usage is very high: always-on video and audio; shared project data and status. The methodology was strongly influenced by agile practices. Work culture elements of the simulated co-location were: cohesive teams (project members in all locations stick together in pursuit of common objectives), and strong common language – in this case English skills.

F. Speed

Given that time zone proximity is an advantage for Brazil, then Follow The Sun (FTS from here on) represents the opposite of Brazil's competitive advantage in its location (with its traditional partners in North America and Europe). FTS has always been the temptress of time zones—always enticing, yet always risky. Brazilian firms, like Indian firms, are looking to try FTS. In 2010, Politec (a large Brazilian IT firm) and Mitsubishi publicized their initiative targeted at Japanese companies to leverage FTS: "The great advantage of such a link between two worlds as seemingly distant is the time zone," said the Japanese executive involved.

The first researchers to examine FTS were Gorton et al [14], where they conducted a series of small controlled experiments. Gupta has also written extensively about the promise of FTS or, more specifically, the 24-hour knowledge factory [15]. During the last decade some have claimed successful FTS practices but while these projects were indeed dispersed, they did not practice the daily hand-offs of FTS [16]. For example, contrary to myth, Indian offshore firms do little FTS [6].

In FTS, each site hands off work at the end of every day to the next site many time zones away (for example, Brazil to India) so that the work can be advanced while one's team rests for the night [9, 12]. Like working two shifts, the two sites reduce duration by half by organizing the tasks to work sequentially on a daily basis. However, complete life cycle FTS is very hard to achieve, and we are not aware of any pure FTS cases. In fact, at one Brazilian site of a large MNC, a serious FTS attempt failed and the project was terminated: FTS couldn't be made to work.

In our study, there were some Brazilian sites that dealt with an Asian location (China or Japan or India). All of these were testing configuration structures in which either the Brazilian location was developing and then the Asian location testing, or vice versa. This arrangement is a classic FTS configuration, though only for the testing phase. However, the Brazilian managers involved in these partnerships did not understand or take advantage of the speed potential from a strategic perspective. For example, at one organization, while much work was being passed back-and-forth frequently between Brazil and India, the work was mediated by a European site and therefore was slowed down and no speed advantages were derived. This is not unusual since most tech managers do not design their work strategically to take advantage of time zones.

Modified agile methodologies may be possible with no overlap, though these contradict the agile principles around interactive communication [12]. When each location uses agile practices and works independently, this is sometimes called Scrum of Scrums (a technique used to scale Scrum for large and, in most cases, distributed teams). This scenario could create an opportunity for FTS (but only if the company fully implements other agile practices such as continuous integration, daily builds, test automation, etc). Only one company in the study had the process culture to implement FTS in such a way, though it did not.

G. Non-overlap Time

The overlap (or timeshift) is rarely a perfect overlap. There is some non-overlap time in almost all the cases of Brazilians working with foreign sites. Opinions are split about whether this is a positive.

Non-overlap time is bad:

- Loss of influence. This appears to be important and was heard at two companies. For example, at one American MNC when dealing with California: "sometimes decisions are made [during non-overlap periods], so we get the perception that we're not involved, we don't have influence."
- Pressure. When it is a European partner, the Brazilians feel pressure to get task/problem resolved before the Europeans go home for the evening.
- Even a small time difference becomes magnified. For example, at one mid-sized Brazilian firm R is the liaison for his projects. "I don't like the 4 hour difference [with Europe]" he says. He points out that there are only 2 hour overlap before he goes to lunch, so as a result, he takes a late lunch to extend his morning overlap.

Non-overlap time is good:

- It is nice to have "quiet time" is a sentiment the authors heard strongly at three Brazilian sites. Quiet time is the period when the Americans don't call or disrupt. "My morning is my quiet time, without calls [from USA]. I do my local meetings in the morning." At another company, where there is no timeshifting, the two-hour time gap with Texas is viewed positively because of the quiet time.
- Small time zone differences can be advantageous in order to reduce distractions. Distraction reduction is especially important when examining workers who are doing cognitively complex work, such as software coding. When the time zone overlap is high, as between (east coast) Americans and Brazilians, there are more distractions on both sides.

All of this leads to the larger question: Is 100% overlap really needed? Does the last 20% of overlap time matter? One manager argued: "We have enough overlap. We're already on flextime. Workers come late. They take calls at home. We've already done much of this. If we do 100% overlap by timeshifting, then retention may become an issue."

H. Unique Time Issues

Brazilian Lunch: Lunch time issues are important all over the world. Brazil is no exception. Generally, Brazilians like to take long lunch breaks, occasionally for as long as two hours. And they like to go out at the same time together as a group/team since lunch is a time to socialize. Contrast this with the American lunch, which tends to be more functional, often for as short as 45 minutes. And many Americans eat lunch at their desk, with very little socializing. As a result of the different cultural understandings of lunch breaks, a number of interviewees recounted variations on the following: Americans would get flustered when they call Brazil and not find anybody from the team at their desk during lunch. This was a revelation for the Brazilian team and forced some adjustments in Brazilian lunchtime behavior.

Holiday Schedules: Brazilians have many holidays. In addition to national holidays, there are statewide and municipal holidays. There are three national secular holidays and eight Christian holidays, including the week-long Carnival. Soccer (football) is the secular religion of Brazil and therefore big games are disruptive to the work schedule. The study was conducted just before the 2010 World Cup.

Equatorial Differences: Equatorial differences are more disruptive than meets the eye. There is the dramatic 2 hour adjustment in daylight savings times that happens twice a year. Additionally, there is the dramatic counter-rhythm that happens with summer vacations since Brazilians have vacations during January and February (summer in Brazil) and Europeans during July and August (summer in Europe). This is best illustrated by a quote from a manager located in Europe that works with his Brazilian subsidiary: “we want to go full speed ahead in January right after the holidays, but in Brazil they are on vacation. And when we are moving slowly in July and August, Brazil is at full speed”.

I. Findings Regarding Location Decisions

We also examined whether a priori location decisions were determined because of Brazil's time zone location. In most cases—91%—the original location decision was not made due to time zone overlap. MNCs come to Brazil for typical emerging market reasons: low cost labor, tax advantages, growth markets, and large pools of human resources. Separately, outsourcing clients choose their Brazilian providers for various reasons, but especially for specific relationships or specific expertise. In some cases, Brazil is a second choice after India, and is picked due to dissatisfaction with the Indian provider. The dissatisfactions are often communications issues that may have stemmed from time zone problems.

The one firm that picked Brazil due to time zones (as a major factor) is an American IT firm. This firm had two large clients that pushed for nearshore / time zone proximate solutions. Client1 asked for Brazil specifically and Client2 asked for Latin America in general. These requests pushed this American company to open the Brazilian site. Several respondents indicated that time zone is a decision factor in the incremental decision of the Brazilian sites — the decision to continue/expand work and strengthen the relationship. The

Brazilian respondents conveyed these narratives: “clients enjoy not doing midnight conf-calls” and “overlap with Brazil is good for the work- life balance [of Americans] because they won't have to get up in the middle of the night.”

The incremental decision interacts with time zones in a virtuous cycle. One senior manager said: “Our turnover in Brazil is low. This is important because it leads to high domain knowledge at our site. Our engineers know the domain. They traveled to [the USA] and then also they build trust and when they talk to [USA] more often [due to time zone] since there is familiarity and trust.”

An American manager explained why he prefers dealing with Brazilian sites: it makes his coordination less structured, whereas with India he sends several e-mails the night before and has to work with more structure. An Indian manager, with experience in multiple continents, mentioned that “Brazil has immediate advantage over India because of time-zone proximity, but Brazilians have to communicate well to [be successful in] this.”

VI. CONCLUSIONS

This paper presented the case of the emerging Brazilian IT industry in order to better understand whether time zone proximity matters for software development. This is a limited sample of Brazilian firms. Sources were mostly Brazilian and therefore were self-interested. Some data were inevitably based on hearsay, e.g., the respondent was not the person who made the location decision about the Brazilian site.

The data collected suggest that software development in Brazil benefits from time zone overlap with North America and Europe. It allows both sides to conduct more of their work interactively rather than as a highly-structured process. This overlap facilitates dense synchronous communications and helps to foster warm relationships between the Brazilians and their foreign partners. The time zone overlap may also help Brazilian sites to obtain and retain unstructured tasks/ projects that tend to be more desirable and profitable.

We note that Brazil's software time zone positioning holds within it a cautionary side. Extreme time zone differences (e.g., US - India) have forced Indian and other global companies to invest in their software development processes in order to work effectively. This is a bitter medicine that has helped some companies in the long run. The convenience of time zone overlap means that some Brazilian companies may continue to be relatively casual from a process perspective.

Is there an ideal location—from a time zone perspective? When India began to emerge as the IT giant, one of its competitive myths was its time zone separation. “we work while you sleep” But Carmel investigated this question vis-à-vis India [7] and found that contrary to myth, the Indian software projects do not benefit from time zone separation. In Brazil we found some support that time zones matter, but not in the mythical sense that we all thought.

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